

Answer Key — Biology Grade 11

Sub-Strand 2.2: Growth and Development in Plants • Lesson 10: Synthesis and Application: Explaining the Farmer's Maize and Beans and Planning the Tree Nursery

Teacher reference only. Do not project or hand out.

Quick marking grid

Q1	Q2	Q3	Q4	Q5	Q6	Q7
D	D	A	B	C	D	D

USE AFTER: PREDICT PHASE — Predict: individual 3-minute prediction on rooting 200 mango and grevillea stem cuttings for a school tree nursery in Machakos

Q1. A school in Machakos wants to grow 200 mango and grevillea seedlings from stem cuttings. Which idea would best help the cuttings form roots?

- A. Remove all the leaves and the cut surface so nothing is left to use water
- B. Keep the cuttings in full sun without water so they become strong
- C. Spread fertiliser on the cut end, because fertiliser makes roots form
- D. Apply a plant hormone source such as crushed bulb onion or Aloe gel to the cut end, because auxins stimulate roots to form**

Correct answer: D. Auxins stimulate root formation on cuttings, and natural sources such as onion and Aloe can supply rooting substances. Fertiliser does not trigger roots, and cuttings need water and some shade.

USE AFTER: OBSERVE PHASE — Observe: Evidence Sorting Station, examining the treated and water-only stem cuttings and the rooting results table

Q2. In a rooting test, four hibiscus cuttings were treated with crushed bulb onion, Aloe gel, cinnamon powder, or water only. Why is the cutting treated with water only included?

- A. It shows which cutting will die first
- B. It is the best treatment because water is the only thing roots need
- C. It makes sure all the cuttings get the same amount of onion
- D. It is the control, used to compare how many roots form with no rooting substance added**

Correct answer: D. The water-only cutting is the control. Comparing it with the treated cuttings shows whether the onion, Aloe or cinnamon caused any change in rooting.

USE AFTER: OBSERVE PHASE — Observe: Evidence Sorting Station, sorting seed factors and dormancy-breaking results (soaking, scarification)

Q3. A farmer plants healthy-looking bean seeds in warm, moist soil with enough air. Most germinate, but a few do not even after many days. What is the most likely reason for the few that fail?

- A. The seeds are dormant, for example with a hard seed coat, so germination has not been triggered**
- B. There is no water in the soil
- C. Beans do not have any hormones
- D. The seeds are monocots, so they cannot germinate

Correct answer: A. Water, air and warmth are present, so the problem lies with the seed. Dormancy, such as a hard seed coat, can stop healthy seeds germinating until it is broken, for example by scarification or soaking.

USE AFTER: EXPLAIN PHASE — Explain: writing the CER final explanation (evidence on hypogeal vs epigeal germination)

Q4. Which statement correctly describes germination in maize and in common bean?

- A. Both are epigeal, because all seeds push their cotyledons above the ground
- B. Maize is hypogeal, with its cotyledon staying underground, while bean is epigeal, with its cotyledons raised above the ground**
- C. Both are hypogeal, because seeds always germinate underground
- D. Maize is epigeal and bean is hypogeal, because maize has two cotyledons

Correct answer: B. In maize (monocot) the cotyledon stays below the soil, which is hypogeal germination. In common bean (dicot) the hypocotyl lifts the cotyledons above the soil, which is epigeal germination.

USE AFTER: EXPLAIN PHASE — Explain: writing the Evidence part of the CER final explanation using class germination data (soaked vs dry beans)

Q5. In a class test, 10 soaked bean seeds and 10 dry bean seeds were kept under the same warm conditions. 8 of the soaked seeds germinated and 3 of the dry seeds germinated. What percentage of the soaked seeds germinated?

- A. 50%
- B. 30%
- C. 80%**
- D. 8%

Correct answer: C. Percentage germination = $8 \div 10 \times 100 = 80\%$. This data is evidence that water helps germination because it activates enzymes and hormones that break down stored food.

USE AFTER: EXPLAIN PHASE — Explain: whole-class check with the Kitale farmer's hybrid maize that failed to germinate in dry soil

Q6. A farmer in Kitale plants hybrid maize seeds in dry soil and they fail to germinate. What is the best advice, and why?

- A. Plant the seeds much deeper, because seeds do not need water to germinate
- B. Add hormones to the dry soil, because hormones make seeds grow without water
- C. Use bigger seeds, because only large seeds need water
- D. Plant when the soil is moist, because water activates enzymes and gibberellins that break down stored food for the embryo**

Correct answer: D. Water is needed for germination: it softens the seed coat and activates enzymes and gibberellins that digest stored food so the embryo can grow. Hormones cannot replace water.

USE AFTER: MODEL BUILDING PHASE — Model: 'Then I thought... Now I know... because...' statements comparing the Day 1 model with the final cumulative model

Q7. A learner writes: 'Then I thought hormones feed a cutting so it can grow. Now I know...' Which ending is correct?

- A. hormones are food stores in the cotyledons that give the plant energy
- B. hormones are only found in animals, so plants use fertiliser instead
- C. hormones are needed in large amounts, like water, to make plants grow
- D. hormones are chemical messengers made in small amounts that control growth, for example auxins stimulate root formation and apical dominance**

Correct answer: D. Plant hormones such as auxins, gibberellins, cytokinins and abscisic acid are chemical messengers that regulate growth in small amounts. They are not food, and plants make them.